

Jupyter Notebook Export

Q1: Write a Python program to sum all the items in a list.

Code

```
l=[int(n) for n in input("enter the list").split(',')]  
print(sum(l))
```

Output

```
enter the list 4,5,67,54,67  
197
```

Q2: Write a Python program to get the largest number from a list.

Code

```
l=[int(n) for n in input("enter the list").split(',')]  
print(max(l))
```

Output

```
enter the list 4,5,6,5,6  
6
```

Q3: Write a Python program to get the smallest number from a list.

Code

```
l=[int(n) for n in input("enter the list").split(',')]  
print(min(l))
```

Output

```
enter the list 4,5,4,2,2  
2
```

Q4: Write a Python program to display the first and last colors from the following list.
color_list = ["Red","Green","White" ,"Black"]

Code

```
ist = ["Red","Green","White" ,"Black"]
print("frist color is"+" "+ist[0],"\n","second color is"+" "+ist[-1] )
```

Output

```
frist color is Red
second color is Black
```

Q5. Write a program that prints minimum and maximum of five numbers entered by the user.

Code

```
l=[int(input("enter the no."+str(i))) for i in range(5)]
l.sort()
print("the mini no. is ",str(l[0]),"    ", "the max no. is ",str(l[-1]))
```

Output

```
enter the no.0 34
enter the no.1 56
enter the no.2 43
enter the no.3 56
enter the no.4 45

the mini no. is  34      the max no. is  56
```

write a Program to check if a string is palindrome or not

Code

```
l=input("enter the string")
if l==l[::-1]:
    print("string is palidrome")
```

Output

```
enter the string 121

string is palidrome
```

Q7: Given a nested list. Write a python program to extend it with adding sub list ["h", "i", "j"] in

a such a way that it will look like the following list

Given List:

list1 = ["a", "b", ["c", ["d", "e", ["f", "g"], "k"], "l"], "m", "n"]

Sub List to be added = ["h", "i", "j"]

Expected output:

['a', 'b', ['c', ['d', 'e', ['f', 'g', 'h', 'i', 'j'], 'k'], 'l'], 'm', 'n']

Code

```
list1 = ["a", "b", ["c", ["d", "e", ["f", "g"], "k"], "l"], "m", "n"]
sub_list = ["h", "i", "j"]
list1[2][1][2].extend(sub_list)
print(list1)
```

Output

```
['a', 'b', ['c', ['d', 'e', ['f', 'g', 'h', 'i', 'j'], 'k'], 'l'], 'm', 'n']
```

Write a python program for Given a Python list, to find value 20 in the list, and if it is present, replace it with 200. Only update the first occurrence of a value

list1 = [5, 10, 15, 20, 25, 50, 20]

Expected output:

list1 = [5, 10, 15, 200, 25, 50, 20]

Code

```
li=[5,10,15,20,25,50,20]
for i in range(len(li)):
    if li[i]==20:
        li[i]=200
        break
print(li)
```

Output

```
[5, 10, 15, 200, 25, 50, 20]
```

Q9. Write a program to generate the sequence: -5, 10, -15, 20, -25..... upto n, where n is an integer input by the user.

Code

```
n=int(input("enter the value"))
l=[t for t in range(-5,-(n+1),-10)]
l2=[t for t in range(10,n+1,10)]
for i in range(len(l)):
    print(l[i],l2[i],end=",")
```

Output

```
enter the value 30
-5 10,-15 20,-25 30,
```

Write a program to find the sum of $1 + \frac{1}{8} + \frac{1}{27} + \dots + \frac{1}{n^3}$, where n is the number input by the user.

Code

```
j=''
n=int(input("enter the no. "))
for i in range(1,n):
    if i**3<=n:
        j=j+'1'+"/"+str(i**3)+'+'
print(j[:-1])
```

Output

```
enter the no. 35
1/1+1/8+1/27
```

Write a program to find the sum of digits of an integer number, input by the user.

Code

```
d=input("enter the no.")
li=[0,]
li=[int(n) for n in d]
print(sum(li))
```

Output

```
enter the no. 3456
18
```

Code

```
j=''
i=65
while j!='Z':
    j=chr(i)
    print(str(i)+"="+str(j))
    i=i+1
```

Output

```
65=A
66=B
67=C
68=D
69=E
70=F
71=G
72=H
73=I
74=J
75=K
76=L
77=M
78=N
79=O
80=P
81=Q
82=R
83=S
84=T
85=U
86=V
87=W
88=X
89=Y
90=Z
```

Q13: Display ascii characters from 48 to 57.

Code

```
j=''
i=48
while i!=58:
    j=chr(i)
    print(str(i)+"="+str(j))
    i=i+1
```

Output

```
48=0
49=1
50=2
51=3
52=4
53=5
54=6
55=7
56=8
57=9
```

Q14: Display the following output with the help of Ascii character.

Code

```
j=''
i=97
while j!='z':
    j=chr(i)
    print(str(i)+"="+str(j))
    i=i+1
```

Output

```
97=a
98=b
99=c
100=d
101=e
102=f
103=g
104=h
105=i
106=j
107=k
108=l
109=m
110=n
111=o
112=p
113=q
114=r
115=s
116=t
117=u
118=v
119=w
120=x
121=y
122=z
```

Q15: Given a Python list. Write a python program to turn every item of a list into its square List1 = [1, 2, 3, 4, 5, 6, 7]

Expected output:

[1, 4, 9, 16, 25, 36, 49]

Code

```
li=[1,2,3,4,5,6,7]
li=[n*n for n in li]
print(li)
```

Output

```
[1, 4, 9, 16, 25, 36, 49]
```

Q16: write a Program to sort alphabetically the words form a string provided by the user. [You can use split() method to split string into a list of words.]

Code

```
l=input("enter the string")
l=list(l)
l.sort()
print("".join(l))
```

Output

```
enter the string mohit
himot
```

Q17: write a Program to Remove Punctuations from a String provided by the user. [Hint: use

punctuation attribute of string module to get all punctuations (i.e. !"#\$%&'\()*+,-./:;<=>?@[\\]^_`{|}~)]

Code

```
import string
user_input = input("Enter a string: ")

punctuations = string.punctuation
result = ""
```

```

for char in user_input:
    if char not in punctuations:
        result += char

print("String without punctuation:", result)

```

Output

```

Enter a string: ert???ty

String without punctuation: erty

```

Q18. Write a menu driven program to develop SMIS that has options to

- accept the marks of the student in five major subjects in Class and display the same.
- calculate the sum of the marks of all subjects. Divide the total marks by number of subjects (i.e. 5), calculate percentage = total marks/5 and display the percentage.

Code

```

while True:
    print("\n----- SMIS MENU -----")
    print("1. Enter Marks")
    print("2. Display Marks")
    print("3. Calculate Percentage")
    print("4. Display Grade")
    print("5. Exit")

    choice = int(input("Enter your choice: "))

    match choice:
        case 1:
            marks = []
            print("Enter marks for 5 subjects:")
            for i in range(5):
                m = float(input(f"Subject {i+1}: "))
                marks.append(m)
            print("Marks saved successfully!")

        case 2:
            if not marks:
                print("No marks entered yet.")
            else:
                print("Marks:", marks)

        case 3:
            if not marks:
                print("Enter marks first.")
            else:
                total = sum(marks)
                percentage = total / 5

```



```

        print("Total Marks:", total)
        print("Percentage:", percentage)

    case 4:
        if not marks:
            print("Enter marks first.")
        else:
            total = sum(marks)
            percentage = total / 5

            if percentage > 85:
                grade = "A"
            elif percentage >= 75:
                grade = "B"
            elif percentage >= 50:
                grade = "C"
            elif percentage >= 30:
                grade = "D"
            else:
                grade = "Reappear"

            print("Percentage:", percentage)
            print("Grade:", grade)

    case 5:
        print("Exiting Program...")
        break

    case _:
        print("Invalid choice. Try again.")

```

Output

```

----- SMIS MENU -----
1. Enter Marks
2. Display Marks
3. Calculate Percentage
4. Display Grade
5. Exit

Enter your choice: 1

Enter marks for 5 subjects:

Subject 1: 34
Subject 2: 5
Subject 3: 5
Subject 4: 4
Subject 5: 3

Marks saved successfully!

----- SMIS MENU -----
1. Enter Marks
2. Display Marks
3. Calculate Percentage
4. Display Grade

```

5. Exit

Enter your choice: 5

Exiting Program...

Q19: Write a python program to print the Following:

```
1
2 1
3 2 1
```

Code

```
for i in range(1,4):
    for j in range(i,0,-1):
        print(j,end=" ")
    print()
```

Output

```
1
2 1
3 2 1
```

Q20: WAP to print the following asterisk pattern:

Code

```
for i in range(1,7):
    for j in range(1,i+1):
        print("*",end=" ")
    print()
```

Output

```
*
* *
* * *
* * * *
* * * * *
* * * * * *
```

Q21: WAP to create a function triangle to print the following asterisk triangle pattern:

Code

```
for i in range(1,7):
    for j in range(1,i+1):
        print("*",end=" ")
    print()
for i in range(5,0,-1):
    for j in range(i,0,-1):
        print("*",end=" ")
    print()
```

Output

```
*
* *
* * *
* * * *
* * * * *
* * * * * *
* * * * *
* * * *
* * *
* *
*
```

Q22: Write a python program to print following multiplication table on the screen

Code

```
for i in range(1,11):
    print(i,"|",end=" ")
    for j in range(1,11):
        print(i*j,end=" ")
    print()
```

Output

```
1 | 1 2 3 4 5 6 7 8 9 10
2 | 2 4 6 8 10 12 14 16 18 20
3 | 3 6 9 12 15 18 21 24 27 30
4 | 4 8 12 16 20 24 28 32 36 40
5 | 5 10 15 20 25 30 35 40 45 50
6 | 6 12 18 24 30 36 42 48 54 60
7 | 7 14 21 28 35 42 49 56 63 70
8 | 8 16 24 32 40 48 56 64 72 80
9 | 9 18 27 36 45 54 63 72 81 90
10 | 10 20 30 40 50 60 70 80 90 100
```

Code

